CS3081 Mini Project — Technical Report (partial)

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Scope statement. This report covers an independent but *incomplete* implementation written in roughly 30 minutes on 2026-11-08. Only items marked "tested" were run, on a Windows laptop with Python 3.10; there was no Docker daemon, k3d, Ollama or student_kit. No training, labelling, deployment or load-test result exists, and none is reported.

Task	Status	Detail
1	not done	Needs the 3,000-command pool, Ollama and a cluster; none available.
2	not done	No labelled data, so no training, losses or scores.
3	implemented, tested (stub)	FastAPI `/parse`, `/healthz`, `/readyz`, headers, schema check, `FAULT_INJECT`; Dockerfile not built.
4	written, not applied	`k8s/app.yaml`: namespace `app`, 2 replicas, requests/limits, probes, Service, Ingress, HPA 1–2 at 70%.
5	done, tested	Subnets and NACL; own evaluator 11/11.
6	partial	Workflow and 9 tests exist; no CI run, no model gate, no `make up`.
7	not done	No deployment to measure.

Task 3 — Inference service

`service/app.py` validates every output against `schema.json` (exactly four keys, eight intents) and returns HTTP 500 with `invalid_model_output` otherwise; `FAULT_INJECT=1` replaces the output by the text `not json`. `/readyz` returns 503 until a background thread has downloaded the model from S3 (boto3, credentials from the environment, i.e. a Secret in Kubernetes). Tests with an injected stub generator cover success and headers, invalid intent, extra key, fault injection and health/readiness (5 of the 9 tests; all pass). **Not verified:** real model loading, the prompt/training format match, image size ($\leq$2 GB), the 50-request CPU/memory test.

Task 4 — Kubernetes

The manifest was written to the specification (namespace `app`, two initial replicas, rolling update, probes, HPA min 1 max 2 target 70%). A unit test checks these fields in the YAML. It was never applied to a cluster, so scaling, rollout, rollback, recovery and cold-start behaviour are **unverified**.

Task 5 — Subnetting and NACL

Subnet	Hosts	Network	Usable range	Usable
app	1,000	10.20.0.0/22	10.20.0.1–10.20.3.254	1,022
public	250	10.20.4.0/24	10.20.4.1–10.20.4.254	254
data	100	10.20.5.0/25	10.20.5.1–10.20.5.126	126
management	20	10.20.5.128/27	10.20.5.129–10.20.5.158	30

Broadcasts: 10.20.3.255, 10.20.4.255, 10.20.5.127, 10.20.5.159. `scripts/subnet_calc.py` reproduces this table and a test asserts it. The NACL (`network/nacl.json`) denies 198.51.100.0/24 as rule 100 (before the web allows 110/120), allows SSH only from 10.20.5.128/27, and has separate reply rules because a NACL is stateless. My own evaluator classified 11 of 11 hand-written packets as expected. Note: the required outbound reply rule (TCP 1024–65535 to anywhere) also permits outbound TCP 8000 to any address. The instructor's `nacl_sim.py` and its six sample packets were **not** run.

Task 6 — CI/CD and AI use

`.github/workflows/ci.yml` runs pyflakes, pytest and gitleaks. **No run has occurred.** The Release-based evaluation, the 95% validity gate and the deliberately failing run are not implemented. `AI_USE.md` states that Claude Code wrote the code and how it was checked.

Limitations and next steps

Obtain the student_kit; label with Phi-3.5, clean, train Pythia-70M, publish a Release; build the image; run on k3d; add the quality gate; run Locust; compute cost and SLO. Until then, Tasks 1, 2, 4 (behaviour), 6 (gate) and 7 have no evidence.